

4. Depict a scene of the effect of kicking a ball in Unity (at 3 various speeds).

FINAL RESULT

On key press:

- Press **1** → **Slow kick**
- Press **2** → **Medium kick**
- Press **3** → **Fast kick**

Ball moves with different speeds using physics.

STEP 1: Create Ground

1. Right-click **Hierarchy** → **3D Object** → **Plane**
 2. Rename → `Ground`
-

STEP 2: Create Ball

1. Right-click → **3D Object** → **Sphere**
2. Rename → `Ball`
3. Set position:

X: 0
Y: 0.5
Z: 0

STEP 3: Add Physics (IMPORTANT)

Select **Ball**:

1. Click **Add Component**
2. Add:
 - **✓ Rigidbody**

Rigidbody settings:

- Mass: 1
- Drag: 0
- Use Gravity: ✓

☞ This allows real movement

🎯 STEP 4: Create Kicker Object

1. Right-click → Create Empty
2. Rename → `Kicker`

Position it slightly behind ball:

X: 0
Y: 0.5
Z: -2

☞ This defines **kick direction**

📖 STEP 5: Create Script

Create script → `BallKick`

Paste this:

```
using UnityEngine;
using UnityEngine.InputSystem;

public class BallKick : MonoBehaviour
{
    public Rigidbody ballRb;
    public Transform kickDirection;

    public float slowForce = 5f;
    public float mediumForce = 15f;
    public float fastForce = 40f;
    public float superFastForce = 80f; // 🖱️ NEW (for key 4)

    void Update()
    {
        if (Keyboard.current.digit1Key.wasPressedThisFrame)
        {
            UnityEngine.Debug.Log("Slow Kick");
            Kick(slowForce);
        }
    }
}
```

```

    if (Keyboard.current.digit2Key.wasPressedThisFrame)
    {
        UnityEngine.Debug.Log("Medium Kick");
        Kick(mediumForce);
    }

    if (Keyboard.current.digit3Key.wasPressedThisFrame)
    {
        UnityEngine.Debug.Log("Fast Kick");
        Kick(fastForce);
    }

    // 🍀 NEW KEY (4)
    if (Keyboard.current.digit4Key.wasPressedThisFrame)
    {
        UnityEngine.Debug.Log("Super Fast Kick");
        Kick(superFastForce);
    }

    // 🟢 Stop key
    if (Keyboard.current.sKey.wasPressedThisFrame)
    {
        StopBall();
    }
}

void Kick(float force)
{
    ballRb.linearVelocity = Vector3.zero;
    ballRb.angularVelocity = Vector3.zero;

    Vector3 dir = (ballRb.transform.position -
kickDirection.position).normalized;

    ballRb.AddForce(dir * force, ForceMode.Impulse);
}

void StopBall()
{
    ballRb.linearVelocity = Vector3.zero;
    ballRb.angularVelocity = Vector3.zero;
}
}

```

🏁 STEP 6: Attach Script

1. Select `Kicker`
 2. Drag `BallKick` script onto it
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🕒 STEP 7: Assign References

Select **Kicker**, in Inspector:

- Ball Rb → drag Ball
 - Kick Direction → drag Kicker
-

► STEP 8: Run

1. Click **Play**
2. Press:

1 → slow
2 → medium
3 → fast
S-> stop

☞ Ball moves with different speeds ⚽